08. Data Analysis

Detailed beginner handbook - English and Arabic

Technologies: Pandas, Polars, PyArrow

What you will learn

Group synthetic data by month using safe tabular-data tools.

Pandas is flexible, Polars is fast for column expressions, and PyArrow provides typed columnar interchange.

الهدف: بناء مثال صغير وفهمه خطوة بخطوة.

شرح بسيط: تعلم الفكرة أولا، ثم جرب المثال الصغير ببيانات اصطناعية فقط.

How to study

Study one lesson at a time. Type examples yourself, predict the result, run the code, then change one thing. Keep a notebook of new words, errors, root causes, and fixes.

Course map

Setup and mental model; ten core concepts; a guided mini-project; the real dashboard architecture; debugging; exercises and answers.

Lesson 1 - Setup and mental model

Prerequisites

Windows 10 or 11, VS Code, PowerShell, and permission to install learning dependencies. Use only synthetic data in tutorials and screenshots.

Setup sequence

1. Open PowerShell.
2. Go to learning-examples/08-data-analysis.
3. Read README.md.
4. Run pip install -r requirements.txt; python analysis.py.
5. Read the complete error if it fails.
6. Change one safe value and rerun.

الخطوات: افتح مجلد المثال، شغل الأمر، غير قيمة واحدة، ولاحظ النتيجة.

Mental model

Treat Pandas, Polars, PyArrow as one layer in a system. Identify the input, the transformation or rule, the output, and possible failures. Understanding that flow is more valuable than memorising syntax.

Checkpoint

Explain: What problem does this technology solve? What is its input? What output should it produce? What can fail?

Lesson 2 - Rows, columns, schemas, data types, nulls, and identifiers

What is it?

A schema names columns and their types. Null means missing, not zero. An identifier labels an entity and should not be accidentally converted to a measurement.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Data Analysis, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
schema = {'Record ID':'string','Month':'string','Total':'integer'}
# Missing is not zero; identifiers are not measurements.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Rows, columns, schemas, data types, nulls, and identifiers.

هذا الدرس يشرح: Rows, columns, schemas, data types, nulls, and identifiers. اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك.

Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 3 - Creating Pandas and Polars DataFrames

What is it?

pd.DataFrame and pl.DataFrame create tabular objects. Inspect shape, column names, dtypes, null counts, and sample rows before calculating anything.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Data Analysis, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
import pandas as pd
import polars as pl
rows={'month':['Jan','Feb'],'total':[12,8]}
pdf=pd.DataFrame(rows); plf=pl.DataFrame(rows)
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Creating Pandas and Polars DataFrames.

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Quick check

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Lesson 4 - Selecting, filtering, sorting, renaming, and deriving columns

What is it?

Select only needed columns, filter with explicit conditions, sort deliberately, rename to canonical names, and derive columns from documented rules.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Data Analysis, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
clean=(plf.select('month','total')
 .filter(pl.col('total')>=0)
 .sort('month')
 .with_columns((pl.col('total')*2).alias('double')))
print(clean)
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Selecting, filtering, sorting, renaming, and deriving columns.

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Lesson 5 - Group-by aggregation, counts, distinct counts, and percentages

What is it?

Group-by partitions rows by key and aggregates each group. Distinct counts require a defined identifier. Percentages need a documented denominator and filter context.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Data Analysis, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
print(plf.group_by('month').agg(
 pl.col('total').sum().alias('total'),
 pl.len().alias('rows')
))
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Group-by aggregation, counts, distinct counts, and percentages.

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Quick check

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Lesson 6 - Joins, relationship keys, duplicates, and unmatched records

What is it?

Joins match keys across tables. Validate expected one-to-one or many-to-one relationships, duplicates, null keys, and unmatched rows to prevent silent multiplication or loss.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Data Analysis, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
people=pl.DataFrame({'id':['A','B'],'name':['One','Two']})
visits=pl.DataFrame({'id':['A','A'],'count':[1,2]})
print(visits.join(people,on='id',how='left'))
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Joins, relationship keys, duplicates, and unmatched records.

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Quick check

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Lesson 7 - Dates, strings, categories, and data cleaning

What is it?

Parse dates with invalid-value handling, normalise text cautiously, preserve raw values when auditing, and use categorical types for controlled repeated labels.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Data Analysis, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
clean=plf.with_columns(
 pl.col('month').str.strip_chars().str.to_uppercase(),
 pl.col('total').cast(pl.Int64, strict=False)
)
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Dates, strings, categories, and data cleaning.

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Quick check

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Lesson 8 - Pandas eager operations and Polars expression style

What is it?

Pandas operations are generally eager and index-aware. Polars builds column expressions and supports lazy optimisation. Translate logic explicitly instead of assuming identical behaviour.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Data Analysis, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
# Pandas
pdf.assign(double=pdf['total']*2)
# Polars expression
plf.with_columns((pl.col('total')*2).alias('double'))
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Pandas eager operations and Polars expression style.

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Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 9 - PyArrow schemas and efficient interchange formats

What is it?

PyArrow stores typed columnar arrays and tables used by Parquet and inter-library transfer. A stable schema prevents repeated guessing and costly conversions.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Data Analysis, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
import pyarrow as pa
table=pa.table({'month':['Jan'],'total':pa.array([12],type=pa.int64())})
print(table.schema)
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates PyArrow schemas and efficient interchange formats.

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Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 10 - Data-quality checks and reproducible transformations

What is it?

Check uniqueness, completeness, validity, consistency, and referential integrity. Make transformation steps deterministic and record rejected or corrected values.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Data Analysis, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
assert plf['month'].null_count()==0
assert plf['total'].min()>=0
assert plf.height==plf.unique().height
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Data-quality checks and reproducible transformations.

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Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 11 - Performance, memory, correctness, and avoiding misleading metrics

What is it?

Optimise only after correctness: select early, avoid row loops, choose efficient types, and profile. Never improve speed by weakening validation or changing metric definitions.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Data Analysis, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
# Select early and avoid Python row loops.
result=(plf.lazy().filter(pl.col('total')>0)
 .group_by('month').agg(pl.col('total').sum()).collect())
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Performance, memory, correctness, and avoiding misleading metrics.

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Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 7 - Guided mini-project

Project goal

Group synthetic data by month using safe tabular-data tools.

Starting example

```
import polars as pl
frame=pl.DataFrame({'month':['Jan','Jan'],'total':[4,8]})
print(frame.group_by('month').agg(pl.col('total').sum()))
```

Read the code

Find imported tools or page elements. Identify the synthetic input. Find the function, event, route, or transformation that changes it. Finally, locate where the result is displayed, returned, saved, or packaged.

Build sequence

1. Run the untouched example.
2. Record the result.
3. Rename one unclear value.
4. Add a synthetic record.
5. Validate empty input.
6. Validate incorrect input.
7. Add a helpful error.
8. Rerun every case.
9. Compare with exercise-solution.
10. Explain every line.

Definition of done

Normal, empty, and invalid cases behave deliberately; no private data is used; and you can explain every line without reading the guide.

Lesson 8 - The real dashboard

Architecture connection

The application combines Pandas, Polars, PyArrow with the rest of the stack. The frontend gathers filters and files; the local API validates requests; the data layer transforms workbook rows; charts present aggregates; export tools create files; and the desktop layer packages the application for Windows.

Trace the upload feature

The user selects a workbook. The frontend sends it to the local API. The backend validates the file and sheets. The data layer creates safe tables. The API returns metadata. React updates state. Plotly displays results. Identify exactly where this guide's technology participates.

Privacy and quality

Do not place narrative, contact, or identifying fields in tutorials, logs, screenshots, tests, or exports. Use generated identifiers and synthetic totals. Validate at each boundary and collect only fields required for the calculation.

Architecture exercise

Draw five boxes: UI, API, Data Processing, Export, Desktop Packaging. Add arrows and label the data. Circle the box related to this guide and add two validation checks.

Lesson 9 - Debugging

Reliable debugging loop

1. Reproduce with the smallest input.
2. Read the first meaningful error.
3. Find the exact file and line.
4. Inspect value and type.
5. Compare expected and actual results.
6. Change one thing.
7. Rerun the same case.
8. Add a regression test.

Common beginner mistakes

- Wrong folder or command.
- Missing dependency or inactive environment.
- Misspelled file, variable, field, route, or import.
- Assuming input is always present.
- Mixing setup, processing, and display in one block.
- Hiding an exception rather than understanding it.
- Using real sensitive data in a test.

أخطاء شائعة: تشغيل الأمر من مجلد خاطئ، نسيان تثبيت التبعيات، أو تغيير أشياء كثيرة بمرة واحدة.

Error notebook

Record: command, expected result, actual error, root cause, smallest fix, and test added. This turns errors into reusable knowledge.

Lesson 10 - Exercises and answers

Exercises

1. Explain the technology in three sentences.
2. Recreate the example without copying.
3. Add one normal synthetic record.
4. Handle empty input.
5. Handle invalid input.
6. Improve unclear names.
7. Add or describe one test.
8. Locate the technology in the dashboard.
9. Record one error and root cause.
10. Add one mini-project feature.

تمرين: أضف شهرا أو قيمة آمنة جديدة، ثم اشرح ما حدث بكلماتك.

Answer guide

A strong solution is observable and explainable: the documented command works; output changes for the new record; empty and invalid cases have deliberate results; names communicate purpose; a test states an expected result; and the architecture explanation identifies the correct boundary.

Next step	Next: 09. Interactive Charts with Plotly